7+1: symbols and their definitions

The symbols of the notebook 7+1.nb and of its documentation, each with its defining equation

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

This is a self-contained dictionary. For each symbol it gives the mathematical notation, the Mathematica name in 7+1.nb, the defining equation and its meaning, and, where it exists, the name used in the earlier notebook 4+4.nb. Sections 2–8 cover the octonions and the Clifford algebras (sections 0–6 of 7+1.nb); section 9 covers the Dirac equation and the Kohn–Sham theories of the field dirac16complex (sections 7–10 of 7+1.nb); section 10 covers gravity, the self-coupled field in gravitational fields and the Standard Model (sections 11–19 of 7+1.nb).

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1. Conventions used in every definition below

- Indices $A, B, K, L \in \{0, 1, \ldots, 7\}$ label the coordinates x_A, the octonion units e_A and the Clifford generators; $i, j \in \{1, \ldots, 7\}$; $h \in \{1, 2, 3\}$; $k \in \{4, \ldots, 7\}$ where stated.
- Mathematica lists are 1-based: the entry η_{AB} of a metric is `η[[A+1, B+1]]`.
- 1_n is the $n \times n$ identity matrix (`IdentityMatrix[n]`, named `ID4`, `ID8`, `ID16`, `ID32`); 0 in a block matrix is a zero block; M^{T} is the transpose, $\overline{M}$ the complex conjugate, $M^{\dagger}$ the conjugate transpose; δ_{AB} is the Kronecker delta; $\{X, Y\} = XY + YX$, $[X, Y] = XY - YX$.
- $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is a block matrix (Mathematica `ArrayFlatten[{{a, b}, {c, d}}]`); $M \otimes N$ is the Kronecker product (`KroneckerProduct[M, N]`).
- An index 8 on a family of eight Clifford generators $G_0, \ldots, G_7$ means the ordered product $G_8 = G_0 G_1 \cdots G_7$ (the convention of 4+4.nb).

- $Cl(p,q)$: p generators square to $+1$ and q to -1 ($v^2 = Q(v)$, Q of signature (p,q)). This label is not the xAct list $\{7,1,0\}$, which declares seven space-like dimensions and one time-like dimension (section 2).

2. Coordinates and metrics

symbol	Mathematica	definition	meaning
X	`X`	$X = (x_0, x_1, \ldots, x_7)$	coordinates; also the components of a symbolic octonion or vector
x_0	`x0`	—	7+1.nb: ordinary time, octonion unit e_0. 4+4.nb: hidden space-like dimension
x_1, x_2, x_3	`x1, x2, x3`	—	ordinary 3-space (both notebooks); octonion units e_1, e_2, e_3
x_4	`x4`	—	7+1.nb: auxiliary space-like, $e_4 = \ell$. 4+4.nb: ordinary time
x_5, x_6, x_7	`x5, x6, x7`	—	7+1.nb: auxiliary space-like, e_5, e_6, e_7. 4+4.nb: auxiliary time-like ("superluminal deflating time")
$\eta^{(4,4)}$	`η4488`	$\mathrm{diag}(1,1,1,1,-1,-1,-1,-1)$	metric of 4+4.nb (space-like $+1$, time-like -1); split-octonion norm
$\eta^{(7,1)}$	`η71`	$\mathrm{diag}(-1,1,1,1,1,1,1,1)$	7+1 metric in the sign convention of 4+4.nb (space-like $+1$): seven space-like and one time-like dimension, the xAct declaration $\{7,1,0\}$; generates $Cl(7,1)$
$\eta^{(1,7)}$	`η17`	$-\eta^{(7,1)} =$ $\mathrm{diag}(1,-1,-1,-1,-1,-1,-1,-1)$	simply the squares of the octonion basis: $\eta^{(1,7)}_{AB} = \mathrm{Re}(e_A e_B)$, $e_0^2 = +1$, $e_i^2 = -1$; $x^{\mathsf{T}}\eta^{(1,7)}x = \mathrm{Re}(xx)$; generates $Cl(1,7)$
$\{7,1,0\}$	`DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`	xAct signature list (section 12 of 7+1.nb, as in Einstein-Lovelock-7+1-Nash.nb)	seven space-like dimensions, one time-like dimension, none degenerate
$\eta^{AB}_{(1,7)}$	`η17inv`	$(\eta^{(1,7)})^{-1} = \eta^{(1,7)}$	inverse metric, raises indices
g^O	`gO`	1_8	norm metric of the octonions, $N(x) = x^{\mathsf{T}} g^O x$
g	`gAB44`	$\eta^{(4,4)}$	norm metric of the split octonions (4+4.nb: `g_AB`)
$\eta^{(5,3)}$	`η53`	$\mathrm{diag}(1,-1,-1,-1,1,1,1,1)$	$\mathrm{Re}(xx)$ of the split octonions

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta^{(7,1)} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta^{(1,7)} = -\eta^{(7,1)} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta^{(1,7)}_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta^{(1,7)}_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

$\eta^{(1,7)}$ gives the signs of the Clifford relation $\Gamma^{(1,7)}_A\Gamma^{(1,7)}_B + \Gamma^{(1,7)}_B\Gamma^{(1,7)}_A = 2\,\eta^{(1,7)}_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta^{(1,7)} = \det\eta^{(7,1)} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta^{(1,7)}_{AB}$ and $\mathrm{Re}(x\,x) = \eta^{(1,7)}_{AB}x^A x^B$.

3. Objects reproduced from 4+4.nb

symbol	Mathematica	definition	meaning (4+4.nb name)
ϵ_{hpq4}	`Qa[h,p,q]`	`Signature[{h, p, q, 4}]`	sign of the permutation (Levi-Civita symbol)
	`Qb[h,p,q]`	$\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4}$	auxiliary symbol
$(s_h)_{pq}$	`s4by4[h]`	$\epsilon_{hpq4} - (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$	self-dual antisymmetric 4×4 (`SelfDualAntiSymmetric`)
$(t_h)_{pq}$	`t4by4[h]`	$\epsilon_{hpq4} + (\delta_{p4}\delta_{qh} - \delta_{ph}\delta_{q4})$	anti-self-dual antisymmetric 4×4 (`AntiSelfDualAntiSymmetric`)
σ	`σ`	$\left(\begin{smallmatrix} 0 & 1_4 \\ 1_4 & 0 \end{smallmatrix}\right)$	split metric on spinor space (`σ`, `σ88`)
ζ_A	`ζ[A]`	$\zeta_0 = 1_8$; $\zeta_h = \left(\begin{smallmatrix} 0 & s_h \\ s_h & 0 \end{smallmatrix}\right)$; $\zeta_{7-h} = \left(\begin{smallmatrix} 0 & t_h \\ -t_h & 0 \end{smallmatrix}\right)$; $\zeta_7 = \zeta_1\cdots\zeta_6$	8×8 building blocks (`ζ[A]`)
τ_A	`τ[A]`	$\tau_A = \zeta_A$	split-octonion matrices (`τ[A]`)
$\bar\tau_A$	`τbar[A]`	$\bar\tau_0 = 1_8$, $\bar\tau_A = \sigma\tau_A^{\mathsf T}\sigma$	conjugate matrices (symbolized tau-bar)
T_A	`T16[A]`	$\left(\begin{smallmatrix} 0 & \bar\tau_A \\ \tau_A & 0 \end{smallmatrix}\right)$, $A = 0,\ldots,7$	16×16 Dirac matrices, $\{T_A, T_B\} = 2\eta^{(4,4)}_{AB}1_{16}$ (`T16^A[A]`)
T_8	`T16[8]`	$T_0T_1\cdots T_7 = \mathrm{diag}(-1_8, 1_8)$	chirality matrix (`T16^A[8]`)
σ_{16}	`σ16`	$T_0T_1T_2T_3 = \mathrm{diag}(-\sigma,\sigma)$	(`σ16`)
P_L, P_R	`PL44`, `PR44`	$\frac12(1_{16} - T_8)$, $\frac12(1_{16} + T_8)$	chiral projectors (`P_L`, `P_R`)
U_h	`uBasis[[h]]`	eigenvectors of σ divided by $\sqrt2$ (order of `Eigensystem[σ]`)	orthonormal eigenbasis of σ (4+4.nb: double-struck u)
	`evalues`, `evecs`	`Eigensystem[σ]`	eigenvalues $(-1,-1,-1,-1,1,1,1,1)$ and eigenvectors of σ
u	`unit`	$U_8 = (1,0,0,0,1,0,0,0)^{\mathsf T}/\sqrt2$, `hUSE = 8`	unit spinor: $\sigma u = u$, $u^{\mathsf T}u = u^{\mathsf T}\sigma u = 1$ (`unit`, `hUSE`)
$E_{44}(h)$	`E44[h]`	$(E_{44})_{Ba} = (U_h^{\mathsf T}\sigma\bar\tau_B)_a$	vielbein (`E^A_a[h]`)
$E_{44}(h)^{-1}$	`E44inv[h]`	$(U_h^{\mathsf T}\sigma U_h)\,(\eta^{(4,4)}_{00}\tau_0U_h, \ldots, \eta^{(4,4)}_{77}\tau_7U_h)$ (columns)	inverse vielbein (`E^a_A[h]`)
ε_B	`eA[B]`, `EA`	$B = 1,\ldots,8$, `EA = Array[eA, 8]`	split-octonion units; $\varepsilon_B \leftrightarrow \tau_{B-1} \leftrightarrow x_{B-1}$
	`gridx44`	`Grid` of $\varepsilon_B\varepsilon_C = \sum_D (g_{BB}E_{44}\tau_{B-1}E_{44}^{-1})_{DC}\varepsilon_D$	split-octonion multiplication table (`gridx`)
	`gridx44From4p4nb`	the table printed by 4+4.nb, Out[325]	reference for the comparison check
L^S_A	`LS[A]`	$g_{AA}\,E_{44}(8)\,\tau_A\,E_{44}(8)^{-1}$	left multiplication by ε_{A+1}
$x \star_S y$	`sMul[x, y]`	$\left(\sum_A x_A L^S_A\right)y$	split-octonion product of coefficient lists
$\bar x$, $N_S(x)$	`sConj`, `sNorm2`	$(x_0, -x_1, \ldots, -x_7)$; $x^{\mathsf T}\eta^{(4,4)}x$	split conjugate and norm
	`zeroDivisor`	$(\varepsilon_1 + \varepsilon_5)(\varepsilon_1 - \varepsilon_5) = 0$	a product of two nonzero split octonions

4. The octonions

symbol	Mathematica	definition	meaning
τ_A^O	`τO[A]`	$\tau_0^O = 1_8$; $\tau_A^O = \tau_A$ $(A = 1,2,3)$; $\tau_A^O = \tau_A\sigma$ $(A = 4,\dots,7)$	octonion 8×8 matrices
$\bar\tau_A^O$	`τObar[A]`	$g^O(\tau_A^O)^{\mathsf T} g^O = (\tau_A^O)^{\mathsf T}$	conjugate matrices; $\tau_A^O\bar\tau_B^O + \tau_B^O\bar\tau_A^O = 2g_{AB}^O 1_8$
E^O	`EO`	$(E^O)_{Ba} = (u^{\mathsf T} g^O \bar\tau_B^O)_a$; $E^O = \eta^{(4,4)} E_{44}(8)$	octonion vielbein (orthogonal)
$(E^O)^{-1}$	`EOinv`	$(E^O)^{\mathsf T} = (u^{\mathsf T} g^O u)(g_{00}^O \tau_0^O u, \dots, g_{77}^O \tau_7^O u)$	inverse vielbein
e_A	`e[A]`, `eList`	$A = 0,\dots,7$, `eList = Array[e, 8, 0]`	octonion units, $e_0 = 1$; $e_A \leftrightarrow \tau_A^O \leftrightarrow x_A$
L_A^O	`LO[A]`	$g_{AA}^O\, E^O \tau_A^O (E^O)^{-1}$	left multiplication by e_A: $e_A e_K = \sum_L (L_A^O)_{LK} e_L$
	`gridO`	`Grid` of $e_A e_K$, same formula as `gridx`	octonion multiplication table
	`gridOExpected`	table computed by `tools/ref71.py`	reference for the comparison check
xy	`oMul[x, y]`	$\left(\sum_A x_A L_A^O\right) y$	octonion product of coefficient lists
$\bar x$	`oConj[x]`	$(x_0, -x_1, \dots, -x_7)$	conjugate
$N(x)$	`oNorm2[x]`	$x^{\mathsf T} g^O x = x_0^2 + \cdots + x_7^2$	norm
$\mathrm{Re}\, x$	`oRe[x]`	x_0	real part
x^{-1}	`oInv[x]`	$\bar x / N(x)$	inverse
$[x,y,z]$	`oAssoc[x, y, z]`	$(xy)z - x(yz)$	associator
	`toE[x]`, `ev[A]`	$\sum_A x_A e_A$; `UnitVector[8, A+1]`	display; coefficient list of e_A
	`xs`, `ys`, `zs`	`Array[a, 8, 0]`, `Array[b, 8, 0]`, `Array[c, 8, 0]`	general symbolic octonions
	`fanoLines`, `orientedLines`	$\{i,j,k\}$ with $e_i e_j = \pm e_k$; (a,b,c) with $e_a e_b = +e_c$	the 7 Fano lines
	`fanoPicture`	centre e_1; vertices e_2, e_4, e_7; midpoints e_6, e_3, e_5	picture of the Fano plane
	`nonAssociativeTriples`	basis triples with $[e_A, e_B, e_K] \neq 0$	there are 168
	`xNum`, `yNum`, `zNum`	$e_0 + 2e_1 - e_4 + 3e_7$, $-2e_0 + e_2 + e_5$, $e_3 + e_6$	numerical example
ℓ	`ev[4]`	$\ell = e_4$	Cayley–Dickson doubling unit, $\ell^2 = -1$
pq, $\bar p$	`qMul`, `qConj`	product in $\mathrm{span}\{e_0,\dots,e_3\}$; $(p_0, -p_1, -p_2, -p_3)$	quaternions $(e_1 e_2 = -e_3)$
(a,b)	`asPair[x]`	$x = a + \ell b$, $a = (x_0,\dots,x_3)$, $b = (x_4,\dots,x_7)$	octonion as a pair of quaternions
	`cdMul[γ]`	$(a,b)(c,d) = (ac + \gamma\, d\bar b,\ \bar a d + cb)$	Cayley–Dickson; $\gamma = -1$ octonions, $+1$ split
	`cdPrompt`, `pm`	$(ac - \bar d b,\ \bar a d + cb)$	the formula of [1A] (fails)
	`promptX`, `promptY`	$e_0 + e_4 = (1,1)$, $e_1 = (i,0)$	its counterexample
s_{AB}	`sAB[A, B]`	-1 if $A \geq 4$ and $B \geq 4$, else $+1$	octonion product $= s_{AB}\times$ split product
c, φ	`cc`, `φ`	$c = (1,1,1,1,i,i,i,i)$, $\varphi(x)_A = c_A x_A$	isomorphism split$\otimes\mathbb{C} \to \mathbb{O}\otimes\mathbb{C}$

5. Clifford algebras

symbol	Mathematica	definition	meaning
T_A^O	`T16O[A]`	$\begin{pmatrix} 0 & \bar\tau_A^O \\ \tau_A^O & 0 \end{pmatrix}$	$Cl(8,0)$: $\{T_A^O, T_B^O\} = 2\delta_{AB} 1_{16}$
T_8^O	`T16O[8]`	$T_0^O \cdots T_7^O = T_8 = \mathrm{diag}(-1_8, 1_8)$	volume element of $Cl(8,0)$
S_{44}	`S44`	$T_4 T_5 T_6 T_7 = \mathrm{diag}(\sigma,\sigma)$	$T_k^O = T_k S_{44}$ for $k \geq 4$
$\Gamma_A^{(1,7)}$	`Γ17[A]`	$\Gamma_0^{(1,7)} = T_0^O = T_0$; $\Gamma_i^{(1,7)} = T_0^O T_i^O$ $(= T_0 T_i,\ i \leq 3$; $= T_0 T_i S_{44},\ i \geq 4)$	REAL 7+1 generators: $\{\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}\} = 2\eta_{AB}^{(1,7)} 1_{16}$
$\Gamma_8^{(1,7)}$	`Γ17[8]`	$\Gamma_0^{(1,7)} \cdots \Gamma_7^{(1,7)} = \begin{pmatrix} 0 & -1_8 \\ 1_8 & 0 \end{pmatrix}$	volume element, square -1_{16}
	`products256`	$\Gamma_{A_1}^{(1,7)} \cdots \Gamma_{A_k}^{(1,7)}$, $A_1 < \cdots < A_k$	the 256 basis products
τ_A^C, $\bar\tau_A^C$	`τC[A]`, `τCbar[A]`	$\tau_0^C = \bar\tau_0^C = i1_8$; $\tau_k^C = \tau_k^O$, $\bar\tau_k^C = \bar\tau_k^O$	complexified 8×8 matrices
$\Gamma_A^{(7,1)}$	`Γ71[A]`	$\begin{pmatrix} 0 & \bar\tau_A^C \\ \tau_A^C & 0 \end{pmatrix}$; $\Gamma_0^{(7,1)} = iT_0$, $\Gamma_k^{(7,1)} = T_k^O$	COMPLEX generators: $\{\Gamma_A^{(7,1)}, \Gamma_B^{(7,1)}\} = 2\eta_{AB}^{(7,1)} 1_{16}$
$\Gamma_8^{(7,1)}$	`Γ71[8]`	$\Gamma_0^{(7,1)} \cdots \Gamma_7^{(7,1)} = iT_8$	volume element, square -1_{16}

symbol	Mathematica	definition	meaning
S	`Ssim`	$(1_{16} - i\Gamma_0^{(1,7)})/\sqrt{2}$	unitary; $\Gamma_A^{(7,1)} = S(i\Gamma_A^{(1,7)})S^{-1}$
	`intertwiners[gs]`	`NullSpace` basis of $\{M : M\overline{g} = gM\ \forall g\}$	Schur intertwiners
	`indicator[gs]`	sign of the scalar $M\overline{M}$	$+1$: $\mathbb{R}(16)$; -1: $\mathbb{H}(8)$
M_0	`M71`	intertwiners of $\Gamma^{(7,1)}$, spanned by $\Gamma_8^{(1,7)}$	$M_0\overline{M_0} = -1_{16}$
ε	`ε2`	$\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$	$\varepsilon^2 = -1_2$
$\widehat{\Gamma}_A^{(7,1)}$	`Γ71R32[A]`	$\varepsilon \otimes \Gamma_A^{(1,7)}$	REAL 32×32 generators with $\eta^{(7,1)}$
Q_i, Q_j, Q_k	`Qi`, `Qj`, `Qk`	$\varepsilon \otimes 1_{16}$, $\sigma_1 \otimes \Gamma_8^{(1,7)}$, $\sigma_3 \otimes \Gamma_8^{(1,7)}$	quaternion units commuting with all $\widehat{\Gamma}_A^{(7,1)}$
	`commutantBasis`, `commutantTime`	`NullSpace` of $\{X : X\widehat{\Gamma}_A^{(7,1)} = \widehat{\Gamma}_A^{(7,1)}X\}$; its timing	dimension 4: $\mathbb{H}$
$\chi^{(7,1)}$	`χ71`	$-i\Gamma_8^{(7,1)} = T_8$	chirality matrix ($\Gamma^{(7,1)}$ basis)
	`PL71`, `PR71`	$\frac{1}{2}(1_{16} \mp \chi^{(7,1)}) = P_L, P_R$	7+1 chiral projectors
$\chi^{(1,7)}$	`χ17`, `PL17`, `PR17`	$i\Gamma_8^{(1,7)}$; $\frac{1}{2}(1_{16} \mp \chi^{(1,7)})$	complex chirality and projectors (rank 8)
E_{16}^O	`EO16`	$\mathrm{diag}(E^O, E^O)$	rotation to the octonion basis
Γ_A^e	`Γ17e[A]`	$E_{16}^O\Gamma_A^{(1,7)}(E_{16}^O)^{\mathsf{T}}$; $\Gamma_0^e = \left(\begin{smallmatrix} 0 & 1_8 \\ 1_8 & 0 \end{smallmatrix}\right)$, $\Gamma_i^e = \mathrm{diag}(L_i^O, -L_i^O)$	$\Gamma^{(1,7)}$ in the octonion basis
$\vec{x}$, ψ	`xImag`, `ψs`	$(0, x_1, \ldots, x_7)$; `Array[ψ, 8, 0]`	imaginary part; a general octonion

6. Examples and bookkeeping

symbol	Mathematica	definition	meaning
Σ_{AB}	`Σ17[A, B]`, `pairs`	$\frac{1}{4}[\Gamma_A^{(1,7)}, \Gamma_B^{(1,7)}]$; `pairs` lists the 28 pairs $A < B$	generators of Spin(1,7)
$S(\beta)^{\pm1}$	`Sβ`, `Sβinv`	$\cosh\frac{\beta}{2}\,1_{16} \pm \sinh\frac{\beta}{2}\,\Gamma_0^{(1,7)}\Gamma_1^{(1,7)}$	boost along x_1 and its inverse
$R(\theta)^{\pm1}$	`Rθ`, `Rθinv`	$\cos\frac{\theta}{2}\,1_{16} \pm \sin\frac{\theta}{2}\,\Gamma_4^{(1,7)}\Gamma_5^{(1,7)}$	rotation in the x_4–x_5 plane and its inverse
Λ	`Λboost`, `Λrot`	$\Lambda_{BK} = \frac{1}{16}\mathrm{tr}(\eta_{BB}^{(1,7)}\Gamma_B^{(1,7)}S\Gamma_K^{(1,7)}S^{-1})$	Lorentz matrices of the two examples
Γ^A	`Γ17up[A]`	$\eta_{(1,7)}^{AA}\Gamma_A^{(1,7)}$	raised index
Ψ	`Ψfun`	$(f_1(X), \ldots, f_{16}(X))$, `f[k] @@ X`	16 arbitrary functions of $x_0, \ldots, x_7$
$\Gamma^A\partial_A$	`DiracOp17[ψ]`	$\sum_A \Gamma^A\,\partial\psi/\partial x_A$	7+1 Dirac operator
$\Box$	`waveOp17[ψ]`	$\sum_A \eta_{(1,7)}^{AA}\,\partial^2\psi/\partial x_A^2$	7+1 wave operator
p, m	`pvec`, `mass`	$p = (3,2,2,0,0,0,0,0)$ (contravariant p^A), $m = 1$	$p^{\mathsf{T}}\eta^{(1,7)}p = m^2$
$\not{p}$	`pslash`	$\sum_A p^A\Gamma_A^{(1,7)}$	$\not{p}^2 = m^2 1_{16}$
W	`realAmplitudes`	`NullSpace[pslash - mass ID16]`	eight real spinor amplitudes, $(\not{p} - m)w = 0$
$p \cdot x$	`pvec.η17.X`	$p^{\mathsf{T}}\eta^{(1,7)}x$	phase of the plane waves
	`Ψwave`	$w\,e^{-i\,p\cdot x}$	solves $(i\Gamma^A\partial_A - m)\Psi = 0$; complex
J	`antiCommutant`	`NullSpace` of $J\Gamma_A^{(1,7)} + \Gamma_A^{(1,7)}J = 0$ $(A = 0, \ldots, 7)$	spanned by $\Gamma_8^{(1,7)}$ alone, $J^2 < 0$
k, $\not{k}$	`kvec`, `kslash`	$k = (1,1,0,0,0,0,0,0)$, $\not{k} = \sum_A k^A\Gamma_A^{(1,7)}$	null momentum, $\not{k}^2 = 0$
	`realNull`	`NullSpace[kslash]`	8 real solutions of $\not{k}\,w = 0$
	`ΨmasslessReal`	$w\cos(k \cdot x)$	REAL solution of $\Gamma^A\partial_A\Psi = 0$
$\eta_{(7,1)}^{AB}$	`η71inv`	$(\eta^{(7,1)})^{-1} = \eta^{(7,1)}$	raises indices for $\widehat{\Gamma}$
$\widehat{\Gamma}^A\partial_A$	`DiracOp71R32[ψ]`	$\sum_A \eta_{(7,1)}^{AA}\widehat{\Gamma}_A^{(7,1)}\,\partial\psi/\partial x_A$	real 32-component Dirac operator
	`Ψfun32`	$(g_1(X), \ldots, g_{32}(X))$, `g32[k] @@ X`	32 arbitrary functions
P	`pslash32`	$\sum_A p^A\widehat{\Gamma}_A^{(7,1)}$	$P^2 = -m^2\,1_{32}$
w	`w32`	`Array[w, 32]`	32 arbitrary real constants

symbol	Mathematica	definition	meaning
	`Ψ32`	$w\cos(p\cdot x)+\frac{1}{m}Pw\sin(p\cdot x)$	REAL solution of $(\widehat{\Gamma}^A\partial_A-m)\Psi=0$
$\Gamma_A^{(5,3)}$	`Γ53[A]`	$\Gamma_0^{(5,3)}=T_0$, $\Gamma_i^{(5,3)}=T_0T_i$	real $Cl(5,3)$ with $\eta^{(5,3)}$
$\Gamma_A^{(0,8)}$	`Γ08[A]`	$T_8^OT_A^O$	real $Cl(0,8)$
	`gens[p]`	T_A^O for $A<p$, iT_A^O for $A\geq p$	complex generators of signature $(p,8-p)$
	`classification`, `claimed1B`, `correct`	$(p,q,\text{indicator})$, $p=8,\ldots,0$; claimed and computed classes	classification table
	`checks`, `check[name, test]`	association of results; stores `TrueQ[test]` and returns `name -> result`	verification bookkeeping
	`failedChecks`, `allChecksPassed`	names of checks that are not True; `AllTrue[Values[checks], TrueQ]`	must be `{}` and `True`

7. Explicit matrices

The 4×4 blocks, the seven split matrices τ_A, and the four new octonion matrices ($\tau_A^O=\tau_A$ for $A\leq 3$):

$$s_1=\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&-1&0&0\\-1&0&0&0\end{pmatrix},\ s_2=\begin{pmatrix}0&0&-1&0\\0&0&0&1\\1&0&0&0\\0&-1&0&0\end{pmatrix},\ s_3=\begin{pmatrix}0&1&0&0\\-1&0&0&0\\0&0&0&1\\0&0&-1&0\end{pmatrix},$$

$$t_1=\begin{pmatrix}0&0&0&-1\\0&0&1&0\\0&-1&0&0\\1&0&0&0\end{pmatrix},\ t_2=\begin{pmatrix}0&0&-1&0\\0&0&0&-1\\1&0&0&0\\0&1&0&0\end{pmatrix},\ t_3=\begin{pmatrix}0&1&0&0\\-1&0&0&0\\0&0&0&-1\\0&0&1&0\end{pmatrix},$$

$$\tau_1=\begin{pmatrix}0&0&0&0&0&0&0&1\\0&0&0&0&0&0&1&0\\0&0&0&0&0&-1&0&0\\0&0&0&0&-1&0&0&0\\0&0&0&1&0&0&0&0\\0&0&1&0&0&0&0&0\\0&-1&0&0&0&0&0&0\\-1&0&0&0&0&0&0&0\end{pmatrix},\quad \tau_2=\begin{pmatrix}0&0&0&0&0&0&-1&0\\0&0&0&0&0&0&0&1\\0&0&0&0&1&0&0&0\\0&0&0&0&0&-1&0&0\\0&0&-1&0&0&0&0&0\\0&0&0&1&0&0&0&0\\1&0&0&0&0&0&0&0\\0&-1&0&0&0&0&0&0\end{pmatrix},$$

$$\tau_3=\begin{pmatrix}0&0&0&0&0&1&0&0\\0&0&0&0&-1&0&0&0\\0&0&0&0&0&0&0&1\\0&0&0&0&0&0&-1&0\\0&1&0&0&0&0&0&0\\-1&0&0&0&0&0&0&0\\0&0&0&1&0&0&0&0\\0&0&-1&0&0&0&0&0\end{pmatrix},\quad \tau_4=\begin{pmatrix}0&0&0&0&0&1&0&0\\0&0&0&0&-1&0&0&0\\0&0&0&0&0&0&0&-1\\0&0&0&0&0&0&1&0\\0&-1&0&0&0&0&0&0\\1&0&0&0&0&0&0&0\\0&0&0&1&0&0&0&0\\0&0&-1&0&0&0&0&0\end{pmatrix},$$

$$\tau_5=\begin{pmatrix}0&0&0&0&0&0&-1&0\\0&0&0&0&0&0&0&-1\\0&0&0&0&1&0&0&0\\0&0&0&0&0&1&0&0\\0&0&1&0&0&0&0&0\\0&0&0&1&0&0&0&0\\-1&0&0&0&0&0&0&0\\0&-1&0&0&0&0&0&0\end{pmatrix},\quad \tau_6=\begin{pmatrix}0&0&0&0&0&0&0&-1\\0&0&0&0&0&0&1&0\\0&0&0&0&0&-1&0&0\\0&0&0&0&1&0&0&0\\0&0&0&1&0&0&0&0\\0&0&-1&0&0&0&0&0\\0&1&0&0&0&0&0&0\\-1&0&0&0&0&0&0&0\end{pmatrix},$$

$$\tau_7 = \begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix},$$

$$\tau_4^O = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}, \quad \tau_5^O = \begin{pmatrix} 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \end{pmatrix},$$

$$\tau_6^O = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}, \quad \tau_7^O = \begin{pmatrix} 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

8. Old names (4+4.nb) and new names (7+1.nb)

4+4.nb	**7+1.nb**	**reason**
σ, σ88	σ	unchanged
τ[A], ζ[A], s4by4[h], t4by4[h]	unchanged	—
symbolized tau-bar	τbar[A]	2-D symbol made with the Notation package; a plain name needs no package
T16^A[A1] (symbolized)	T16[A]	the superscript looked like a power and the letter A was not an index
P_L, P_R	PL44, PR44	the 7+1 projectors get their own names PL71, PR71, PL17, PR17
double-struck u	uBasis	plain name
g_AB	gAB44	subscripted name; plain name
E^A_a[h], E^a_A[h] (symbolized)	E44[h], E44inv[h]	the super- and subscripts were part of the name, not indices
eA[1..8] (1-based)	e[0..7] (0-based) for the octonions	aligns e_A with τ_A^O and x_A: no $B-1$ shift
gridx	gridx44, and the new grid0	two tables now

9. Sections 7–10 of 7+1.nb: the field dirac16complex and its Kohn–Sham theories

The symbols used for the Dirac equation and the Kohn–Sham theories of sections 7–10 of 7+1.nb. $\Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A^{(1,7)}$, $\Gamma_8^{(1,7)}$ and $\chi^{(1,7)}$ are those of section 5 of this dictionary; here Γ_A means $\Gamma_A^{(1,7)}$, $\Gamma^0 = \Gamma_0^{(1,7)}$, and $\chi = \chi^{(1,7)}$.

symbol	Mathematica	definition	meaning
Ψ, Ψ_a	`dirac16complex`, `ψ16[a]`	$\Psi = (\Psi_1, \ldots, \Psi_{16})^{\mathsf{T}}$, `Table[ψ16[a] @@ X, {a, 1, 16}]`	quantized 16-component field; complex anticommuting operators
$\Psi_\uparrow$, $\Psi_\downarrow$	—	$\Psi = (\Psi_\uparrow; \Psi_\downarrow)$	upper and lower 8 components
$\bar\Psi$	—	$\Psi^\dagger\Gamma^0$	Dirac adjoint
Ψ^c	—	$\Gamma_8^{(1,7)}\Psi^*$	charge conjugate; $(\Psi^c)^c = -\Psi$
$\delta^7(\vec x - \vec y)$	—	$\int \delta^7(\vec x - \vec y) f(\vec y)\, d^7y = f(\vec x)$	Dirac delta in the 7 space coordinates $\vec x = (x_1, \ldots, x_7)$
m	`mD`	$m > 0$	mass; $m = 1$ in the examples
$H_{\rm D}$	`hD[ψ, m]`	$-i\Gamma^0\Gamma^i\partial_i + m\Gamma^0$	Dirac Hamiltonian; Hermitian
$\vec p$, p^i, E	`pv7`, `Ep`	$\vec p = (p^1, \ldots, p^7)$, $E = \sqrt{m^2 + \vec p^{\,2}}$	momentum and energy of a plane wave $e^{-i(Ex_0 - \vec p\cdot\vec x)}$
e_s	`UnitVector[8, s]`	$(e_s)_r = \delta_{rs}$	unit column of length 8
w_s	`uS[s, M, E]`	$\frac{1}{\sqrt{E}}\big(E\, e_s;\ (m1_8 - p^i\tau_i^O)e_s\big)$	real positive-energy amplitudes, $(\not p - m)w_s = 0$, $\not p = E\Gamma_0 + p^i\Gamma_i$
$P_L^{(1,7)}$, $P_R^{(1,7)}$	`PL17`, `PR17`	$\frac12(1_{16} \mp \chi^{(1,7)})$	complex chiral projections
Ψ_L, Ψ_R	`ϕL`, `ϕR`	$\frac{\Psi_\uparrow + i\Psi_\downarrow}{\sqrt2}$, $\frac{i\Psi_\uparrow + \Psi_\downarrow}{\sqrt2}$	chiral 8-component fields
g_S, g_P, g_V, g_A	`gS`, `gP`, `gV`, `gA`	real constants	scalar, pseudoscalar, vector, axial-vector contact couplings (mass dimension -6)
$\mathsf{s}, \mathsf{p}, \mathsf{v}_A, \mathsf{a}_A$	—	real functions of x	external sources
$\mathcal{L}$	—	below	Lagrangian density of the self-coupled field
ρ_S, ρ_P	`dens16[M]`	$\bar\Psi\Psi$, $\bar\Psi i\chi\Psi$	scalar and pseudoscalar densities
j^A, j_5^A	`dens16[M]`	$\bar\Psi\Gamma^A\Psi$, $\bar\Psi\Gamma^A\chi\Psi$; $j_A = \eta^{(1,7)}_{AA} j^A$	vector and axial-vector currents
Φ_k, λ_k	`orbV`	$H_{\rm KS}\Phi_k = \lambda_k\Phi_k$, $\int \Phi_k^\dagger\Phi_l = \delta_{kl}$	Kohn–Sham orbitals and eigenvalues
$H_{\rm KS}$	`hKS[p]`	below	Kohn–Sham Hamiltonian
$\mathcal{S}, \mathcal{P}, \mathcal{V}_A, \mathcal{A}_A$	`Sks`, `Pks`, `VKS`, `AKS`	below	Kohn–Sham potentials (lower index A)
F	—	$F[n] = \min_{\Psi\to n}\langle\Psi\vert H_0 + H_{\rm int}\vert\Psi\rangle$	universal functional of the 18 densities $n = (\rho_S, \rho_P, j^A, j_5^A)$
T_s	—	$\sum_k \int \Phi_k^\dagger H_{\rm D}\Phi_k\, d^7x$	kinetic plus rest energy of the orbitals
E_H	`EH16`	below	Hartree energy
E_{xc}, E_x	`Exfun[R]`	$E_{xc} = F - T_s - E_H$; E_x below	exchange-correlation and exact exchange energies
$\mathcal{R}$	`Rorb`	$\sum_k \Phi_k\bar\Phi_k$	16×16 density matrix
Φ_{kL}, Φ_{kR}	`orbLV`, `orbRV`	$\frac{\Phi_{k\uparrow} + i\Phi_{k\downarrow}}{\sqrt2}$, $\frac{i\Phi_{k\uparrow} + \Phi_{k\downarrow}}{\sqrt2}$	chiral components of the orbitals
κ	`κc`	$\sum_k \Phi_{kL}^\dagger\Phi_{kR}$	complex scalar density: $\rho_S = \kappa + \kappa^*$, $\rho_P = i(\kappa - \kappa^*)$
$\mathcal{M}$	`Mks`	$m + \mathcal{S} + i\mathcal{P}$	complex Kohn–Sham mass
$\mathcal{V}_A^L$, $\mathcal{V}_A^R$	`VLks`, `VRks`	$\mathcal{V}_A - \mathcal{A}_A$, $\mathcal{V}_A + \mathcal{A}_A$	chiral vector potentials
ϑ	`Uθ`	$e^{i\vartheta\chi} = \cos\vartheta\, 1_{16} - \sin\vartheta\,\Gamma_8^{(1,7)}$	chiral rotation angle; `Uθ` has $\cos\vartheta = \frac35$
k_F, m^*	`kF0`, `gap7`	$m^* = m + g_S\rho_S(m^*)$	Fermi momentum and effective mass of uniform matter
ρ	`rho7`	$j^0 = k_F^7/(105\pi^4)$	particle density of uniform matter
E/A, E_F	`eps7/rho7`, `fermi7`	$E_F = g_V\rho + \sqrt{k_F^2 + m^{*2}}$	energy per particle and Fermi energy of uniform matter
$\Lambda_{\rm c}$	`Λc`	$\vert\vec p\vert < \Lambda_{\rm c}$, $\Lambda_{\rm c} = 1$	momentum cutoff of the Dirac sea
g, $g_{\rm c}$	`gNJL`, `gcrit`	$g = g_S = g_P$; $g_{\rm c} = -90\pi^4/\Lambda_{\rm c}^6$	chirally symmetric coupling; critical coupling

Defining equations:

$$\big(i\Gamma^A\partial_A - m\big)\Psi = 0, \qquad \big(i\partial_0 + \tau_i^O\partial_i\big)\Psi_L = m\Psi_R, \qquad \big(i\partial_0 - \tau_i^O\partial_i\big)\Psi_R = m\Psi_L,$$

$$\mathcal{L} = \bar\Psi\big(i\Gamma^A\partial_A - m\big)\Psi - \bar\Psi\big(\mathsf{s} + i\chi\mathsf{p} + \Gamma^A(\mathsf{v}_A + \chi\mathsf{a}_A)\big)\Psi$$
$$\quad - \frac{g_S}{2}(\bar\Psi\Psi)^2 - \frac{g_P}{2}(\bar\Psi i\chi\Psi)^2 - \frac{g_V}{2}(\bar\Psi\Gamma^A\Psi)(\bar\Psi\Gamma_A\Psi) - \frac{g_A}{2}(\bar\Psi\Gamma^A\chi\Psi)(\bar\Psi\Gamma_A\chi\Psi),$$
$$H_{\rm KS} = \Gamma^0\Gamma^i\big(-i\partial_i + \mathcal{V}_i\big) + \Gamma^0(m + \mathcal{S}) + \mathcal{V}_0 + \Gamma^0 i\chi\mathcal{P} + \chi\mathcal{A}_0 + \Gamma^0\Gamma^i\chi\mathcal{A}_i,$$
$$\mathcal{S} = \mathsf{s} + g_S\rho_S + \frac{\delta E_{xc}}{\delta\rho_S},\quad \mathcal{P} = \mathsf{p} + g_P\rho_P + \frac{\delta E_{xc}}{\delta\rho_P},\quad \mathcal{V}_A = \mathsf{v}_A + g_V j_A + \frac{\delta E_{xc}}{\delta j^A},\quad \mathcal{A}_A = \mathsf{a}_A + g_A j_{5A} + \frac{\delta E_{xc}}{\delta j_5^A},$$
$$E_H = \int\Big[\frac{g_S}{2}\rho_S^2 + \frac{g_P}{2}\rho_P^2 + \frac{g_V}{2}j^A j_A + \frac{g_A}{2}j_5^A j_{5A}\Big]d^7x,$$
$$E_x = -\frac{g_S}{2}\mathrm{tr}(\mathcal{R}\mathcal{R}) - \frac{g_P}{2}\mathrm{tr}(i\chi\mathcal{R}\,i\chi\mathcal{R}) - \frac{g_V}{2}\mathrm{tr}(\Gamma^A\mathcal{R}\Gamma_A\mathcal{R}) - \frac{g_A}{2}\mathrm{tr}(\Gamma^A\chi\mathcal{R}\Gamma_A\chi\mathcal{R}),$$
$$\big[\mathcal{V}_0^L - \tau_i^O\partial_i - i\tau_i^O\mathcal{V}_i^L\big]\Phi_L + \mathcal{M}\Phi_R = \lambda\Phi_L,\qquad \mathcal{M}^*\Phi_L + \big[\mathcal{V}_0^R + \tau_i^O\partial_i + i\tau_i^O\mathcal{V}_i^R\big]\Phi_R = \lambda\Phi_R.$$

In $H_0 + H_{\rm int}$, $H_0 = \int :\Psi^\dagger H_{\rm D}\Psi: d^7x$ and $H_{\rm int} = \int\big[\frac{g_S}{2}:\rho_S^2: + \frac{g_P}{2}:\rho_P^2: + \frac{g_V}{2}:j^A j_A: + \frac{g_A}{2}:j_5^A j_{5A}:\big]d^7x$ (normal ordered); $\min_{\Psi\to n}$ runs over the states with densities n.

10. Sections 11–19 of 7+1.nb: gravity, the self-coupled field in gravitational fields, the Standard Model

The symbols of the third stage of the project (sections 11–19 of 7+1.nb). Here $\Gamma_A = \Gamma_A^{(1,7)}$, $\Gamma^A = \eta^{AA}_{(1,7)}\Gamma_A$, $\eta = \eta^{(1,7)}$, and Greek indices $\mu,\nu,\lambda,\rho,\sigma,\alpha,\beta,\gamma,\delta \in \{0,\dots,7\}$ are coordinate indices (on the brane $\mu,\nu \in \{0,\dots,3\}$).

symbol	Mathematica	definition	meaning
$g_{\mu\nu}$, $g^{\mu\nu}$	`cv["g"]`, `cv["ginv"]`	one time-like, seven space-like dimensions: xAct signature $\{7,1,0\}$; components with the signs of η ($g_{00} > 0$)	metric of 7+1 spacetime and its inverse
$\{^\lambda_{\mu\nu}\}$	`cv["chr"]`	(10.1)	Christoffel symbols
$R^\rho{}_{\sigma\mu\nu}$	`cv["riem"]`	(10.1)	Riemann tensor
$R_{\sigma\nu}$, R, $G^\mu{}_\nu$	`cv["ric"]`, `cv["R"]`, `cv["G"]`	(10.1)	Ricci tensor, scalar curvature, Einstein tensor
$R^{\gamma\delta}{}_{\alpha\beta}$	`cv["rud"]`	$g^{\delta\sigma}R^\gamma{}_{\sigma\alpha\beta}$	Riemann tensor, two indices up
$\delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n}$	`gKD`, `kδ`	$\det[\delta^{\mu_i}_{\nu_j}]$	generalized Kronecker delta
$E^{(k)\mu}{}_\nu$, L_k	`Lovelock1/2/3`, `lovelockE`, `lovelockL`	(10.2)	Lovelock tensors and densities
α_k, $\bar\Lambda$, κ, G_8	`alf[k]`, `lamb`, `kap`	(10.3)	couplings, cosmological constant, $\kappa = 8\pi G_8$
w_k, Λ, H	`w1`, `w2`, `w3`, `Λ`, `H`	(10.3)	pure-number couplings and inverse length of the reference notebook
$T^\mu{}_\nu$; ε, p_a, p_b	—; `ksStress`	$T = \mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, \dots, -p_b)$	energy–momentum tensor; energy density and pressures
a, b, H_a, H_b	`a3[x0]`, `b4[x0]`, `Ha`, `Hb`	(10.4); $H_a = a'/a$, $H_b = b'/b$	scale factors and Hubble rates
q_H	`qH`	$b = e^{q_H H x_0}$ when $a = e^{Hx_0}$	ratio of the Hubble rates of exponential solutions
p, q	`pK`, `qK`	$a = x_0^p$, $b = x_0^q$	Kasner exponents
$e^A{}_\mu$, $e_A{}^\mu$	`fd["e"]`, `fd["ei"]`	$g_{\mu\nu} = \eta_{AB}e^A{}_\mu e^B{}_\nu$	frame field and its inverse
$\Gamma_\mu(X)$, $\Gamma^\mu(X)$	`fd["Gam"]`, `fd["GamUp"]`	$e^A{}_\mu\Gamma_A$, $e_A{}^\mu\Gamma^A$	generalized frame matrices
$\omega_\mu{}^A{}_B$, Ω_μ	`fd["om"]`, `fd["Om"]`	(10.5)	canonical spin connection, spinor connection
D_μ, ∇_μ	—	$D_\mu = \partial_\mu + \Omega_\mu$; (10.5)	spinor covariant derivative; total covariant derivative
$R_{AB\mu\nu}$	`rABmn`	$\eta_{AC}e^C{}_\rho R^\rho{}_{\sigma\mu\nu}e_B{}^\sigma$	Riemann tensor with frame indices
q, A_μ	`qG`, `aGauge`	$D_\mu = \partial_\mu + \Omega_\mu - iqA_\mu$	charge and abelian gauge field
M (pair creation), $\lvert\beta\rvert^2$	`bogoliubov`	$\lvert\beta\rvert^2 = 1/(e^{2\pi M/H}+1)$ in de Sitter space	mass of a mode; pair-creation probability
K, Θ^K, Θ_K	`mIdx`, `0up[K]`, `0dn[K]`	(10.6)	multi-index and the 256 Clifford products
g_r; g_S, g_V, g_A, g_P	`gg[r]`; `gS`, `gV`, `gA`, `gP`	$g_0 = g_S$, $g_1 = g_V$, $g_7 = -g_A$, $g_8 = -g_P$	self-couplings of rank r; the four of section 11.3

symbol	Mathematica	definition	meaning
F_{rs}, G_s	`fierzF`, `Geff`	(10.6)	Fierz matrix and Hartree–Fock couplings
M (Kohn–Sham), ρ_S, ρ	`ksGap`, `ksMoments`	$M = m + G_0\rho_S$	effective mass, scalar density, density of uniform matter
W, $h_{\mu\nu}$, L_c, b_0	—	$W = \{x_4 = \cdots = x_7 = 0\}$, $h_{\mu\nu} = g_{\mu\nu}(x, y = 0)$	world brane, induced metric, torus period, static scale
S, $S_{\rm EL}$, S_Ψ, $S_{\rm stab}$	—	(10.9)	total action; Einstein–Lovelock, dirac16complex and stabilizing parts
$\mathcal{L}_{\rm self}$, $\mathcal{L}_{\rm SM}$, $\mathcal{L}_{\rm portal}$	`LselfS`	(10.9); $\mathcal{L}_{\rm self} = -\frac{1}{2}\sum_r g_r I_r$	self-coupling, Standard Model and portal Lagrangians
$V_\perp$, G_4	`vPerp`	$V_\perp = b_0^4 L_c^4$, $G_4 = G_8/V_\perp$	proper volume of the torus; four-dimensional Newton constant
v, G_F, J	`vEW`, `gFermi`, `jCKM`	$v = (\sqrt{2}G_F)^{-1/2}$; $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$	Higgs vacuum value, Fermi constant, Jarlskog invariant
$V_{\rm CKM}$, $U_{\rm PMNS}$, U	`vCKM`, `uPMNS`	standard parametrization	quark and lepton mixing matrices
ρ_4, Λ_4	—	$3H_a^2 = 8\pi G_4\rho_4 + \Lambda_4$	energy density and cosmological constant of the brane
γ_μ, γ_5, ρ_k	`g4D`, `g5D`, `rhoK`	(10.7)	brane Dirac matrices and transverse matrices
R_A, U (brane)	`rBr`, `uInt`	(10.7)	brane form of the Dirac matrices; intertwiner
ψ_a, e_a	—	$\Psi = U\sum_a \psi_a \otimes e_a$	four brane Dirac spinors; unit vectors of $\mathbb{C}^4$
$\vec{q}$	`qKK`	$\vec{q} = 2\pi\vec{n}/L_c$	auxiliary momentum
μ, m, N	`muPortal`, `mPsi`, `nAct`	$1 - NN^\dagger = \mu\mu^\dagger/m^2$	portal Dirac mass, dirac16complex mass, light mixing matrix
$P(\alpha\to\beta)$, X_{ij}, Δm^2_{ij}	`pVac`, `xFormula`, `dm21`, `dm31`	(10.8)	oscillation probability
L, E, n_e	—		baseline, neutrino energy, electron density

$$\{{}^{\lambda}_{\mu\nu}\} = \tfrac{1}{2}g^{\lambda\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}), \tag{10.1}$$

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu\{{}^{\rho}_{\sigma\nu}\} - \partial_\nu\{{}^{\rho}_{\sigma\mu}\} + \{{}^{\rho}_{\mu\lambda}\}\{{}^{\lambda}_{\sigma\nu}\} - \{{}^{\rho}_{\nu\lambda}\}\{{}^{\lambda}_{\sigma\mu}\},$$

$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \quad G^{\mu}{}_{\nu} = R^{\mu}{}_{\nu} - \tfrac{1}{2}\delta^{\mu}_{\nu}R,$$

$$E^{(k)\mu}{}_{\nu} = -\frac{1}{2^{k+1}}\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \tag{10.2}$$

$$L_k = \frac{1}{2^k}\delta^{\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k},$$

$$0 = -\Lambda\delta^{\mu}_{\nu} + \sum_{k=1}^{3} H^{-2k} w_k E^{(k)\mu}{}_{\nu}, \quad \sum_{k=0}^{3}\alpha_k E^{(k)\mu}{}_{\nu} = \kappa T^{\mu}{}_{\nu}, \tag{10.3}$$

$$\alpha_0 = 2\bar{\Lambda},\ \alpha_1 = 1,\ \alpha_2 = \tfrac{w_2}{w_1 H^2},\ \alpha_3 = \tfrac{w_3}{w_1 H^4},$$

$$ds^2 = dx_0^2 - a^2(dx_1^2 + dx_2^2 + dx_3^2) - b^2(dx_4^2 + \cdots + dx_7^2), \tag{10.4}$$

$$\omega_\mu{}^{A}{}_{B} = e^{A}{}_{\nu}\big(\partial_\mu e_B{}^{\nu} + \{{}^{\nu}_{\mu\lambda}\}e_B{}^{\lambda}\big), \quad \Omega_\mu = \tfrac{1}{4}\eta_{AC}\omega_\mu{}^{C}{}_{B}\Gamma^A\Gamma^B, \tag{10.5}$$

$$\nabla_\mu\Gamma_\nu = \partial_\mu\Gamma_\nu - \{{}^{\lambda}_{\mu\nu}\}\Gamma_\lambda + [\Omega_\mu, \Gamma_\nu] = 0,$$

$$\Theta^K = i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r},\ \ \sigma_r = \tfrac{r(r-1)}{2} \bmod 2,\ \ \Theta_K = \eta_{A_1A_1}\cdots\eta_{A_rA_r}\Theta^K, \tag{10.6}$$

$$F_{rs} = \tfrac{1}{16}\sum_l (-1)^{rs-l}\tbinom{s}{l}\tbinom{8-s}{r-l},\ \ G = (1 - F^{\mathsf{T}})g,$$

$$\gamma_0 = \mathrm{diag}(1,1,-1,-1),\ \gamma_i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix},\ \rho_4 = i\gamma_0,\ \rho_{4+i} = -\gamma_i, \tag{10.7}$$

$$R_\mu = \gamma_\mu \otimes 1_4,\ R_k = \gamma_5 \otimes \rho_k,\ \Gamma_A U = U R_A,$$

$$P(\alpha\to\beta) = \Big|\sum_i U_{\beta i}U_{\alpha i}^* e^{-im_i^2 L/(2E)}\Big|^2, \quad X_{ij} = U_{\alpha i}^* U_{\beta i} U_{\alpha j} U_{\beta j}^*, \quad \Delta m^2_{ij} = m_i^2 - m_j^2, \tag{10.8}$$

$$S = S_{\rm EL} + S_\Psi + \int_W d^4x\sqrt{|h|}(\mathcal{L}_{\rm SM} + \mathcal{L}_{\rm portal}) + S_{\rm stab}, \tag{10.9}$$
$$S_{\rm EL} = -\tfrac{1}{2\kappa}\int d^8X\sqrt{|g|}\sum_k \alpha_k L_k, \quad S_\Psi = \int d^8X\sqrt{|g|}\big[\bar\Psi(i\Gamma^\mu D_\mu - m)\Psi + \mathcal{L}_{\rm self}\big].$$

11. Sections 20–23 of 7+1.nb: the coupled field equations and their solutions

The symbols of the fourth stage (sections 20–23 of 7+1.nb). A prime is d/dx_0; $\tau = x_0 - x_s$ is the time since the singularity; regime I uses units $\ell_2 = \sqrt{\alpha_2} = 1$.

symbol	Mathematica	definition	meaning
L_2, L_3	`feL3coef`	(11.1)	Gauss–Bonnet and cubic Lovelock densities
α_k, Λ, κ	`alf[k]`, `lamb`, `kapS`	$\alpha_0 = 2\Lambda$, $\alpha_1 = 1$, $\kappa = 8\pi G_8$	Lovelock couplings, bulk cosmological constant
ℓ_2, ξ	`rIxi`	$\ell_2 = \sqrt{\alpha_2}$, $\xi = \alpha_3/\alpha_2^2$	Gauss–Bonnet length, cubic ratio
ε, p_a, p_b, $W(a,b)$	`epsM`, `paM`, `pbM`, `wW`	(11.2)	energy density, pressures, comoving energy
$U(b)$, U_0, K_U, b_0	`uPot`, `u0B`, `kU`	$U_0 + \frac{K_U}{2}\ln^2(b/b_0)$	stabilizing potential
β, m_r, ϕ	`betR`, `rIIm`	$\ln(b/b_0)$, $m_r^2 = \kappa K_U/12$, $\phi = \sqrt{12}M_{\rm Pl}\beta$	radion, its mass, canonical radion
f_W, $V_\perp$, G_4, Λ_4	`fWb`, `vPerpS`	$V_\perp = b^4L_c^4$, $8\pi G_4 = \kappa/V_\perp$, $\Lambda_4 = \Lambda + \kappa U_0 + \kappa f_W/V_\perp$	brane tension and four-dimensional constants
u, X, P_E, P_G, P_C	`uS`, `xS`, `rIpE`, `rIpG`, `rIpC`	(11.3)	vacuum curve of regime I
p_1, q_1, p_2, q_2	`rIp1`, `rIq1`, `rIp2`, `rIq2`	(11.3)	Kasner exponents of the Einstein and Gauss–Bonnet regimes
h_0, h_*	`rIcub[[1]]`, `rIhStar`	$-\sqrt{\alpha_2/(30\alpha_3)}$, $-\sqrt{\alpha_2/(54\alpha_3)}$	auxiliary contraction rates near $\tau = 0$ (vacuum, matter)
$p_{\rm eff}$	`rIpeff`	$-H_a^2/H_a'$	effective Kasner exponent
S_1, S_2, S_3, $f_{\rm gas}$	`rIstate[h]["S"]`, `["frac"]`	(11.4)	sizes of the curvature terms; gas fraction
A, ϑ, τ_E	`rIIaF`, `rIIphF`, `rIItEF`	$\beta \approx A\sin(m_r\tau + \vartheta)/(m_r(\tau - \tau_E))$	radion oscillation after capture
g_*, g_{*s}, $t(T)$	`rIIgstars`, `rIItg`	$\rho = \frac{\pi^2}{30}g_*T^4$, $s = \frac{2\pi^2}{45}g_{*s}T^3$	brane plasma
H_∞, γ, $m_{\rm eff}$	`rIIIH`, `rIIIgam`, `rIIImeff`	$H_0\Omega_\Lambda^{1/2}$; (11.5)	de Sitter rate; radion friction and mass

$$L_2 = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \tag{11.1}$$
$$L_3 = \frac{1}{8}\delta^{\alpha_1\beta_1\alpha_2\beta_2\alpha_3\beta_3}_{\gamma_1\delta_1\gamma_2\delta_2\gamma_3\delta_3}R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}R^{\gamma_2\delta_2}{}_{\alpha_2\beta_2}R^{\gamma_3\delta_3}{}_{\alpha_3\beta_3},$$
$$\varepsilon = \frac{W}{a^3b^4}, \quad p_a = -\frac{\partial_a W}{3a^2b^4}, \quad p_b = -\frac{\partial_b W}{4a^3b^3}, \tag{11.2}$$
$$ds^2 = dx_0^2 - a^2d\vec{x}^2 - b^2d\vec{y}^2,$$
$$P_E = 3(2u^2+4u+1), \quad P_G = -12u(u^3+12u^2+18u+4), \quad P_C = 360u^3(3u+4), \tag{11.3}$$
$$u = \frac{H_b}{H_a}, \quad X = H_a^2, \quad p_1 = \frac{1+2\sqrt{2}}{7}, \quad q_1 = \frac{2-3\sqrt{2}}{14}, \quad p_2 = \frac{3}{3+4u_2}, \quad q_2 = u_2p_2,$$
$$S_1 = 3(H_a^2 + 4H_a|H_b| + 2H_b^2), \qquad S_2 = 12\alpha_2|H_b|(4H_a^3 + 18H_a^2|H_b| + 12H_aH_b^2 + |H_b|^3), \tag{11.4}$$
$$S_3 = 360\alpha_3H_a^2|H_b|^3(4H_a + 3|H_b|), \qquad f_{\rm gas} = \frac{\kappa\varepsilon}{S_1+S_2+S_3+\kappa\varepsilon},$$
$$\frac{\gamma}{3H} = \frac{1 - 16\alpha_2H^2 + 48\alpha_2^2H^4 - 48\alpha_3H^4 + 512\alpha_2^3H^6 + 192\alpha_2\alpha_3H^6 - 1024\alpha_2^4H^8}{(1 - 4\alpha_2H^2 + 32\alpha_2^2H^4)^2}, \tag{11.5}$$
$$\frac{m_{\rm eff}^2}{m_r^2} = \frac{1}{1 - 4\alpha_2H^2 + 32\alpha_2^2H^4}.$$